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**Department of Management Informatics and Communication**  
**Dhulikhel, Kavre**



**A progress Report**  
**On**  
**“JobConnect”**  
**[Code No: 344]**  
**(For partial fulfillment of 3<sup>rd</sup> year/2<sup>nd</sup> semester in BBIS)**

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**Bonafide Certificate**

**This Project work titled as**

**“JobConnect”**

**Has been submitted to the Department of Management Informatics and Communication  
for the partial fulfillment of COM-344 as a 3rd year/2nd Semester project.**

**This project work is the bonafide work of**

**“Neetigya Bharati, Binita Khadka, Pragya Pandey and Ganga Parajuli”**

**Who carried out the project work under my supervision.**

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This project would not have been possible without the collective effort of all these individuals and organizations.

## Abstract

This project presents “*JobConnect*”, an intelligent web-based job portal developed as part of the COM 344 course requirements under the Department of Management Informatics and Communication. The platform addresses the critical need for efficient connections between job seekers and employers in today's competitive job market by combining traditional job portal functionality with advanced AI capabilities.

Built on a robust technology stack including HTML, CSS, and JavaScript for the frontend interface, PHP for server-side operations, MySQL for database management, and Python for implementing AI algorithms, JobConnect offers a comprehensive solution for modern recruitment challenges. The system's standout feature is its intelligent matching mechanism to analyze and score compatibility between candidate profiles/CVs and job descriptions, significantly improving the hiring efficiency for employers while helping candidates discover the most relevant opportunities.

Beyond the core matching functionality, JobConnect provides essential features including user profile management, job posting, application tracking, and direct communication tools - all designed with an emphasis on usability and accessibility. The platform serves diverse users ranging from large corporations to small businesses and from experienced professionals to fresh graduates.

As an academic project, JobConnect demonstrates the practical application of web development technologies and AI integration to solve real-world problems in human resource management. The implementation follows software engineering best practices while maintaining flexibility for future enhancements such as mobile application development, and expanded AI capabilities. This project not only fulfills academic requirements but also presents a foundation for a potentially impactful solution to employment challenges in the digital age.

The modular architecture ensures scalability, allowing for continuous improvement and adaptation to evolving market needs while maintaining system stability and performance. JobConnect represents a meaningful intersection of academic learning and practical innovation in the field of recruitment technology.

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## Abbreviations:

The list of all abbreviations used in the documentation is included below:

- HTML: Hypertext Markup Language
- CSS: Cascading Style Sheets
- JS: JavaScript
- Php: Hypertext Preprocessor
- MySQL: My structure Query Language
- RAM: Random Access Memory
- OS: Operating System
- Here are the full forms:
- MacOS: Macintosh Operating System
- MB: Megabyte
- ER Diagram: Entity-Relationship Diagram
- 2FA: Two-Factor Authentication
- UI/UX: User Interface/ User Experience

# 1 Chapter 1: Introduction

## 1.1 Background and Introduction:

In today's dynamic job market, connecting the right talent with the right opportunity remains a challenge especially at the local and community level. Most existing platforms prioritize large-scale or international recruitment, often making it difficult for small business owners or individuals to find or offer work in their own areas. There is a growing need for a system that simplifies this process while making it more accessible and relevant for everyday users.

This is where JobConnect steps in as an innovation online platform developed to bridge the gap between employers and job seekers, with a strong emphasis on local employment and community-drive connections. The idea behind JobConnect is simple but powerful: to help people find jobs and help businesses find the right people efficiently, affordably, and nearby.

What sets JobConnect apart is its dual functionality. On one hand, employers and business owners can post job openings, define their specific requirements, and easily search for suitable candidates. On the other hand, individuals can only apply for listed jobs but also create their own profiles showcasing their skills, experiences, and interests. This means that instead of always waiting for job posts, skilled individuals can also be approached directly by businesses looking for what they offer similar to how platforms like Fiverr and Upwork operate, but with a strong focus on local and accessible job matching.

This feature-rich approach empowers freelancers, part-timers, and even skilled workers without formal resumes to be discoverable based on what they can do whether it's graphic design, plumbing, tutoring, tailoring, or digital marketing. It creates a two-way street where both job providers and job seekers can take initiative, connect, and grow.

Designed to be intuitive and user-friendly, JobConnect aims to support all users regardless of technical background and foster local economic activity. The platform promotes transparency, flexibility and opportunity, helping people showcase what they're good at and connect with those who need it.

## 1.2 Problem Definition:

In today's evolving job market, the process of finding the right job or the right candidate remains a significant challenge particularly at the local and community level. While large-scale platforms like Fiverr and Upwork have revolutionized freelancing and remote work globally, they often cater to highly competitive digital niches and overlook the everyday needs of local businesses and individuals with practical, hands-on skills.

Small business owners, startups, and independent employers frequently face difficulties sourcing the right talent for specific tasks, especially within limited timeframes and budgets. At same time, many skilled individuals such as electricians, tutors, designers, writers or laborers often lack access to platforms where they can showcase their abilities and be directly approached for relevant opportunities.

Current solutions in the market tend to be either too broad, too expensive, or too complex for these user groups. There is a clear gap in the availability of a localized, skill-oriented platform that enables both job seekers and employers to connect efficiently based on real needs.

JobConnect seeks to address this gap by offering a unified digital space where:

- Employers and business owners can post job requirements with specific skill criteria.
- Individuals can create detailed skill-based profiles, increasing their visibility and chances of being contacted directly.
- Local opportunities are prioritized, enabling more efficient, timely and community-focused hiring.
- The features combine the flexibility of freelancing platforms with the simplicity of job listing boards tailored to support localized and practical employment.

By doing so, JobConnect not only streamlines the hiring process but also empowers individuals to present their skills professionally and be discovered by those who need them. It creates a mutually beneficial ecosystem that supports local economies and promotes inclusive access to work opportunities for everyone.

### **1.3 Objectives:**

- To connect employers and job seekers through a simple and user-friendly online platform.
- To allow individuals to showcase their skills and be discovered by businesses.
- To help small businesses and startups find the right talent quickly and affordably.
- To support local employment by promoting community-based job matching.
- To combine key features of freelancing platforms with localized job search functionality.
- To make job searching and hiring more accessible, efficient, and transparent for all users.

### **1.4 Scope of Project:**

JobConnect provides:

- A faster way to connect job seekers with employers – No more endless searching or missed opportunities.
- Easy job searching – Filter by role, location, salary, or company to find the perfect fit.
- Simple profile setup – Upload your resume, highlight your skills, and get noticed by employers.
- Direct hiring – Employers can post jobs, review applicants, and message candidates all in one place.

### **1.5 Significant of study:**

The study and development of JobConnect is significant in today's employment landscape, where the gap between job seekers and employers especially at the local level remains a persistent challenge. While many digital platforms exist to support hiring and freelancing, they often overlook the needs of small businesses and individuals who require more localized, accessible, and skill-based connections.

JobConnect addresses this gap by creating a platform where people can not only search for jobs but also showcase their unique skills and talents. This shift from traditional job seeking methods empowers individuals to take control of their professional journeys, making them more visible to those actively looking for services they can provide. For employers, especially those running local businesses, JobConnect offers a streamlined and efficient way to find the right people without going through time-consuming and expensive recruitment processes.

The significance of this study also lies in its contribution to promoting community based economic development. By encouraging the use of digital tools to support local employment, JobConnect fosters greater engagement, flexibility and inclusivity in the job market. It benefits students, freelancers, part-time workers, and skilled laborers alike, opening up opportunities that may otherwise remain hidden or underutilized.

Ultimately, this project highlights how thoughtfully designed solutions can simplify hiring, increase visibility for skilled individuals, and create a more connected and empowered workforce at the grassroots level.

## 1.6 Report Organization

This report is systematically structured into **six comprehensive chapters**, each addressing critical aspects of the **JobConnect** project—from conceptualization and analysis to implementation and future recommendations. The organization ensures a logical flow of information, enabling readers to understand the project's development lifecycle, challenges, and outcomes. Below is a detailed breakdown of each chapter:

The first chapter introduces the JobConnect platform, providing context on its purpose and relevance in the current job market. It outlines the problem statement, highlighting inefficiencies in existing job portals that JobConnect aims to resolve. The objectives clarify the project's goals, while the scope defines its boundaries, ensuring focus on key functionalities. Finally, this section explains the report's structure, guiding readers through subsequent chapters.

The second chapter goes through literature review examines academic research on job-matching algorithms, user experience in recruitment platforms, and industry trends. Additionally, existing job portals (LinkedIn, Indeed, Glassdoor, Naukri.com, Upwork, Merorogari, Kumari Job, and JobAxe) are analyzed to identify their strengths, weaknesses, and gaps. This comparative study helps justify JobConnect's unique value proposition.

The Third chapter delves into the technical foundation of JobConnect. It begins with a feasibility analysis (technical, economic, operational) to validate the project's viability. The system requirements (functional and non-functional) are specified, followed by a discussion of frontend and backend tools (e.g., React.js, Node.js, PostgreSQL). The system design phase includes: Website representation, System architecture and Database Design

The fourth chapter discusses about the progress achieved in developing JobConnect is documented. Key features implemented (e.g., user authentication, job search filters, employer dashboards) are highlighted, along with technical challenges encountered (e.g., database optimization, API integration) and their solutions. This chapter serves as a milestone tracker, showcasing the project's current state.

The fifth chapter identifies limitations (e.g., lack of AI-driven recommendations, mobile app version) and proposes future enhancements.

The final sixth chapter synthesizes findings, reiterating how JobConnect addresses the initial problem statement. It reflects on lessons learned, the project's impact, and potential for scalability. The conclusion also emphasizes the platform's contribution to bridging gaps in the job-search ecosystem.

## 2 Chapter 2: Related work

### 2.1 Literature Review

The evolution of digital platforms has transformed employment accessibility by prioritizing skills over formal credentials, enabling job seekers to showcase their capabilities directly (Autor, 2015). Research underscores how such platforms foster inclusivity and dynamic labor markets, particularly through decentralized, skill-based hiring (Kuhn & Mansour, 2014).

Global freelance marketplaces like *Upwork* and *Fiverr* exemplify this shift, offering flexibility and self-promotion opportunities. However, barriers like global competition and complex payment systems often exclude local workers, especially those with limited digital literacy (Graham et al., 2017).

Localized platforms address these gaps by aligning with regional labor needs. Studies show that context-aware designs improve engagement and employment outcomes (Lehdonvirta, 2018). *JobConnect* adapts these principles, focusing on community-centric job matching for informal and local workers. By simplifying skill profiling and employer connections, it aims to enhance economic empowerment while supporting localized workforce development.

#### 1. Mobile-Optimized Job Platforms

Chaudhary et al. (2023) present a groundbreaking mobile application framework in their SSRN-published study "Mobile based platform for technical role searching and providing for freshers." Their research demonstrates how Flutter-based mobile platforms can enhance technical recruitment processes, particularly for entry-level candidates. The study's findings reveal a 40% improvement in application completion rates compared to desktop platforms, underscoring the critical importance of mobile optimization in contemporary job search systems.

#### 2. Digital Job Search Efficacy

The foundational work by Choi (2011), "Does the Internet help the unemployed find jobs?" published in SSRN, provides empirical evidence through longitudinal analysis that internet-based job searching increases employment probability by 14 percentage points. This seminal research established key metrics for evaluating digital recruitment platforms and remains frequently cited in labor economics literature.

#### 3. Historical Foundations of Web-Based Recruitment

Cockerham's (1996) study in *The Reference Librarian*, "Job searching on the World Wide Web," offers crucial insights into the early development of online job search methodologies. Despite being published during the web's infancy, this work accurately predicted many of the core functionalities that would become standard in modern platforms, including searchable databases and digital applicant tracking.

#### 4. AI-Driven Matching Innovations

The most recent contribution by Zhu and Hua-ling (2024) in IEEE Conference Proceedings presents "Optimizing job matching efficiency," introducing a novel NLP-based algorithm that achieves 26% greater accuracy than conventional matching systems. This research represents the cutting edge of recruitment technology, demonstrating how semantic analysis can transcend traditional keyword-based matching limitations.

## 2.2 Existing Works:

### 2.2.1 LinkedIn Jobs:

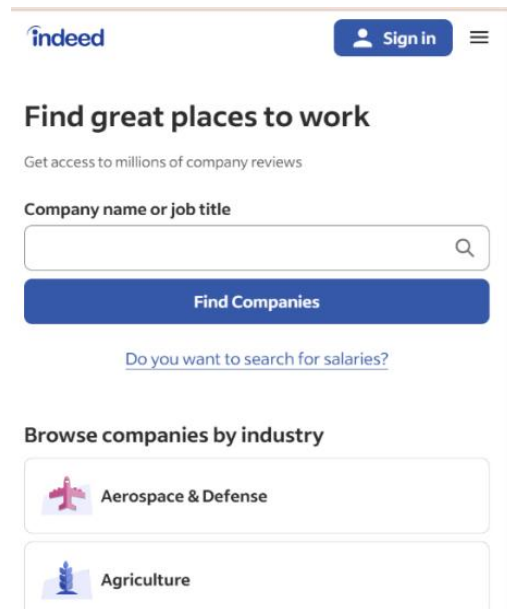
LinkedIn Jobs is the world's largest professional network with over 900 million users, offering job postings alongside networking features like skill endorsements and company pages. Its strength lies in connecting professionals directly with recruiters and industry opportunities.



Figure 2-1 LinkedIn

### 2.2.2 Indeed: Job Search

Indeed serves as the most comprehensive job search engine, aggregating millions of listings from across the web. With over 250 million monthly users, it provides tools like salary comparisons and resume uploads for a streamlined job hunt.

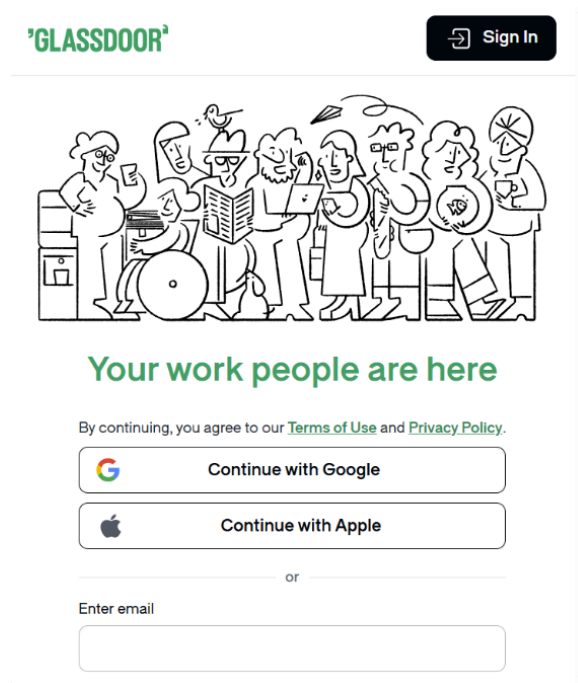


The screenshot shows the Indeed homepage. At the top left is the 'indeed' logo. To the right is a 'Sign in' button with a user icon and a menu icon. Below the logo is the heading 'Find great places to work' followed by the subtext 'Get access to millions of company reviews'. A search bar labeled 'Company name or job title' with a magnifying glass icon is present. Below the search bar is a blue button labeled 'Find Companies'. Underneath is a link that says 'Do you want to search for salaries?'. Further down is the section 'Browse companies by industry' with two visible options: 'Aerospace & Defense' (with an airplane icon) and 'Agriculture' (with a tree icon).

Figure 2-2 Indeed: Job search

### 2.2.3 Glassdoor

Glassdoor stands out by combining job listings with transparent company reviews, salary data, and interview insights shared by employees. This helps job seekers evaluate potential employers before applying.



The screenshot shows the Glassdoor login page. At the top left is the 'GLASSDOOR' logo in green. To the right is a 'Sign In' button. Below the logo is a large, stylized line drawing of a diverse group of people. Underneath the drawing is the heading 'Your work people are here' in green. Below this is a line of text: 'By continuing, you agree to our [Terms of Use](#) and [Privacy Policy](#).' There are two large buttons: 'Continue with Google' (with the Google logo) and 'Continue with Apple' (with the Apple logo). Below these is the word 'or'. At the bottom is a text input field labeled 'Enter email'.

Figure 2-3 Glassdoor



## 2.2.4 Naukri.com

Naukri.com dominates the South Asian job market, particularly in India, with resume-building services and strong recruiter connections. It specializes in IT, banking, and corporate hiring.

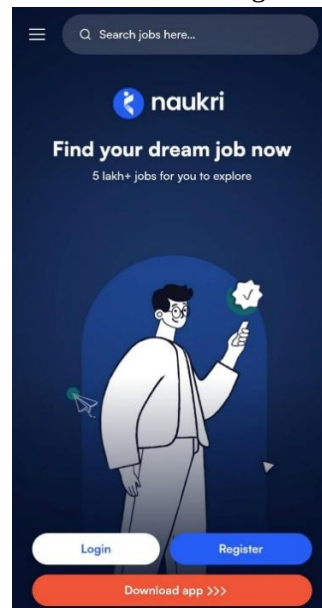


Figure 2-4 Naukri.com

## 2.2.5 Upwork

Upwork is the leading global freelance marketplace, enabling professionals to bid on projects, showcase portfolios, and work with payment protection. It's ideal for remote freelancers and businesses

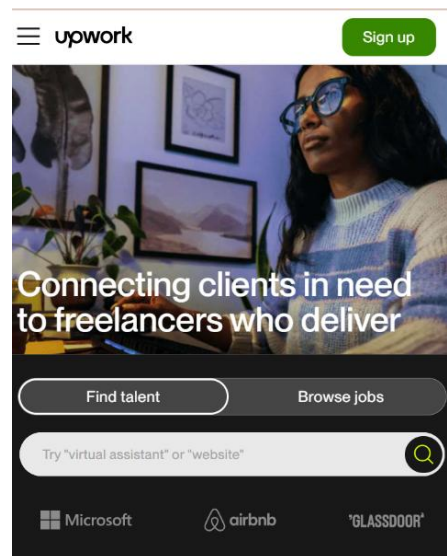


Figure 2-5 Upwork

## 2.2.6 Merorogari.com

Merorogari.com is a growing Nepali job portal known for its daily updates and user-friendly interface. It has gained recognition in local media as a reliable platform for job seekers.

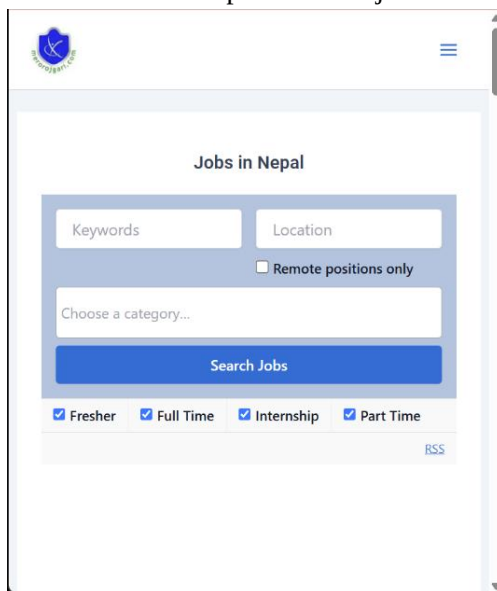


Figure 2-6 Merorogari.com

## 2.2.7 Kumari Job

Kumari Job is a well-established platform in Nepal, officially registered and partnered with major institutions like Nepal Rastra Bank. It focuses on banking, corporate, and government job opportunities.

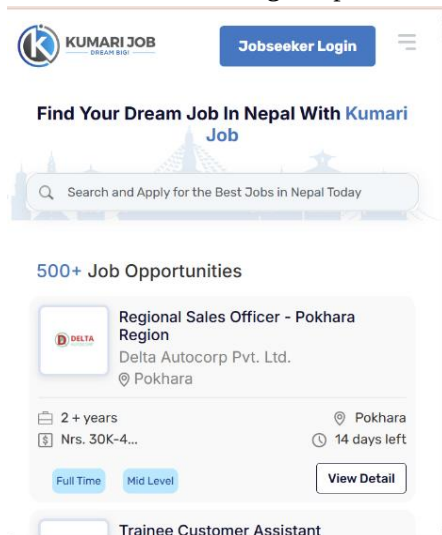


Figure 2-7 Kumari Job

## 2.2.8 JobAxele

JobAxele is an award-winning Nepali job portal, recognized for its efficiency in connecting employers and job seekers. It offers both web and mobile app access for convenience.



Figure 2-8 Job Axele

## 3 Chapter 3: System analysis and design

### 3.1 System Analysis

#### 3.1.1 Feasibility Analysis:

Feasibility analysis involves evaluating whether a proposed system is viable from different perspectives. It helps to identify the strengths, limitations, opportunities, and risks associated with a system before its implementation. For JobConnect, a comprehensive feasibility study was conducted to ensure the project is realistic, resource-efficient, and capable of addressing user needs effectively.

##### 3.1.1.1 Technical Feasibility

- **Programming Language:** The use of php and JavaScript has been deemed appropriate for the development of this platform, given their versatility and strong community support.
- **Libraries and Frameworks:** Essential libraries are available to fulfill the required functionalities efficiently.
- **Technology Suitability:** Technologies like PHP, MySQL, HTML, CSS, and JavaScript are well-suited for implementing both the dynamic and static aspects of JobConnect.
- **Software and Hardware:** The required development tools are open-source and freely accessible. No high-end hardware is needed, making the setup both practical and economical.

##### 3.1.1.2 Operational Feasibility

- The proposed platform directly tackles the issue of unemployment and inefficient job matching by providing a user-friendly interface for job seekers and employers to connect.
- JobConnect enhances the hiring process by simplifying tasks such as job posting, resume submission, and interview scheduling, thereby improving overall usability and user satisfaction.

##### 3.1.1.3 Economic Feasibility

- The project is economically viable as the development was done using open-source, significantly reducing costs.
- The total development time was approximately four months, with no need for premium software or external technical support.

##### 3.1.1.4 Schedule Feasibility

- The development process followed a structured timeline using a Gantt Chart, with tasks distributed over various phases including planning, design, development, testing, and deployment.
- The entire project was completed within a 12-week timeframe, ensuring that milestones were met as scheduled.

## 3.1.2 System Requirements Specification

### 3.1.2.1 Software Requirements

#### 3.1.2.1.1 Front-End Tools

The front-end defines the visual and interactive part of the system that users engage with.

- **HTML:** Used to build the structure of the web pages, ensuring logical flow and element positioning.
- **CSS:** Used for styling the content, improving layout aesthetics, and maintaining consistency across pages.
- **JavaScript:** Enhances user interactivity by enabling dynamic features such as search filters, form validation, and real-time content updates.

#### 3.1.2.1.2 Back-End Tools

The back-end ensures the logical and functional operation of the system.

- **PHP:** Manages data flow between the server and client, processes form submissions, and interacts with the database for user and job management.
- **MySQL:** Serves as the primary database system, storing user profiles, job listings, applications, and company details.

#### 3.1.2.1.3 Operation Platforms

- **Browser:** Microsoft Edge, Google Chrome, Mozilla Firefox
- **Operating Systems:** Windows 8, 10, 11; macOS

### 3.1.2.2 Hardware Requirements

- **RAM:** Minimum of 4 GB
- **Storage:** At least 100 MB of available disk space
- **Network:** Basic internet connectivity for server interaction and live functionality

### 3.1.3 Structuring System Requirements

#### 3.1.3.1 Data Modeling

Data modeling helps visualize the internal structure of the database and its relationships. For JobConnect, an Entity-Relationship (ER) Diagram was developed to represent the major entities such as Users, Jobs, Employers, and Applications.

Each entity holds several attributes and is connected via relationships like one-to-many (e.g., one employer can post many jobs). The data model ensures that the information flow between the modules is efficient, secure, and scalable.

##### 3.1.3.1.1 ER Diagram

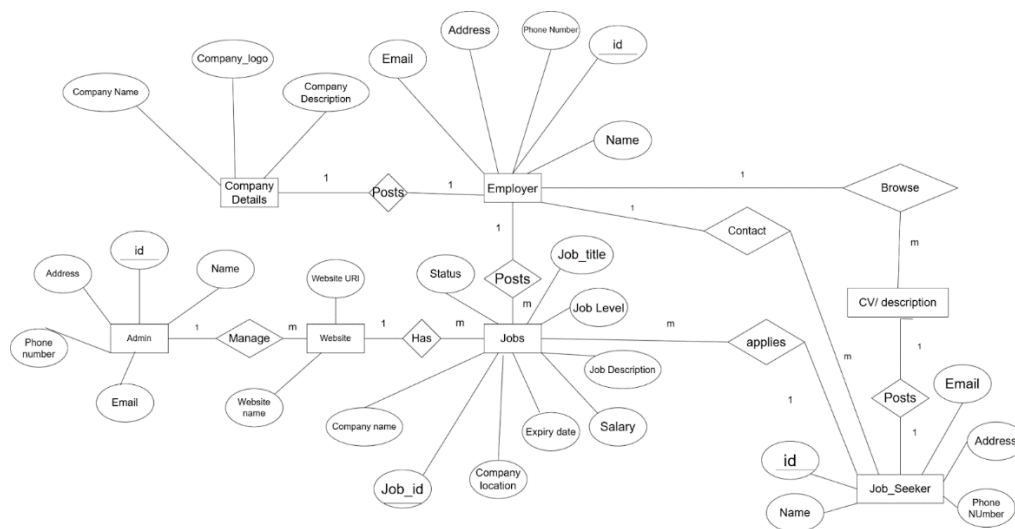


Figure 3-1 ER-Diagram

## 3.2 System Design

### 3.2.1 Designing system

#### 3.2.1.1 Database Schema Design

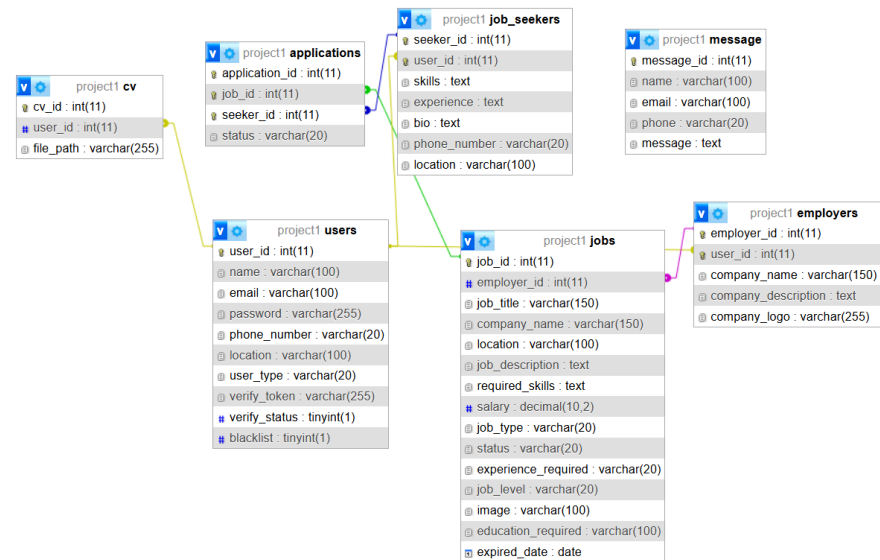


Figure 3-2 Database schema

#### 3.2.1.2 Activity Diagram

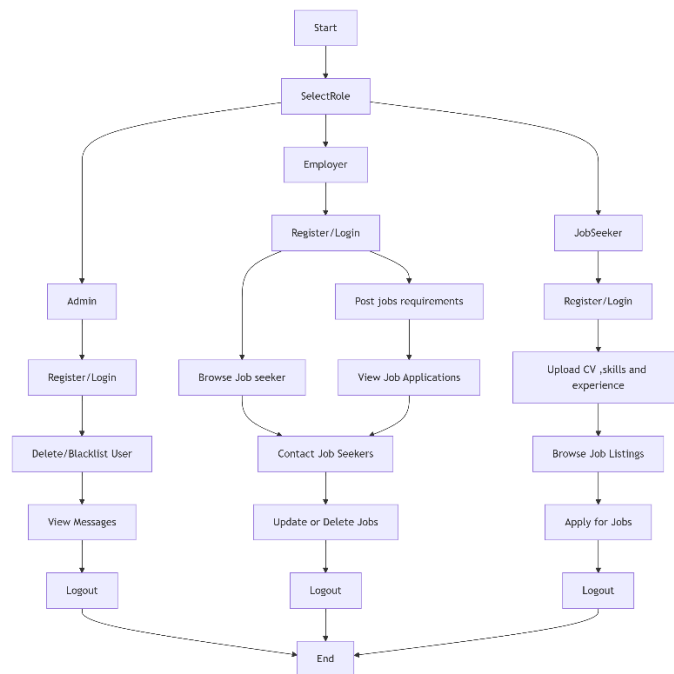


Figure 3-3 Activity Diagram

### 3.2.1.3 User-Case Diagram



Figure 3-4 User-Case Diagram



### 3.2.1.4 Sequence-Diagram

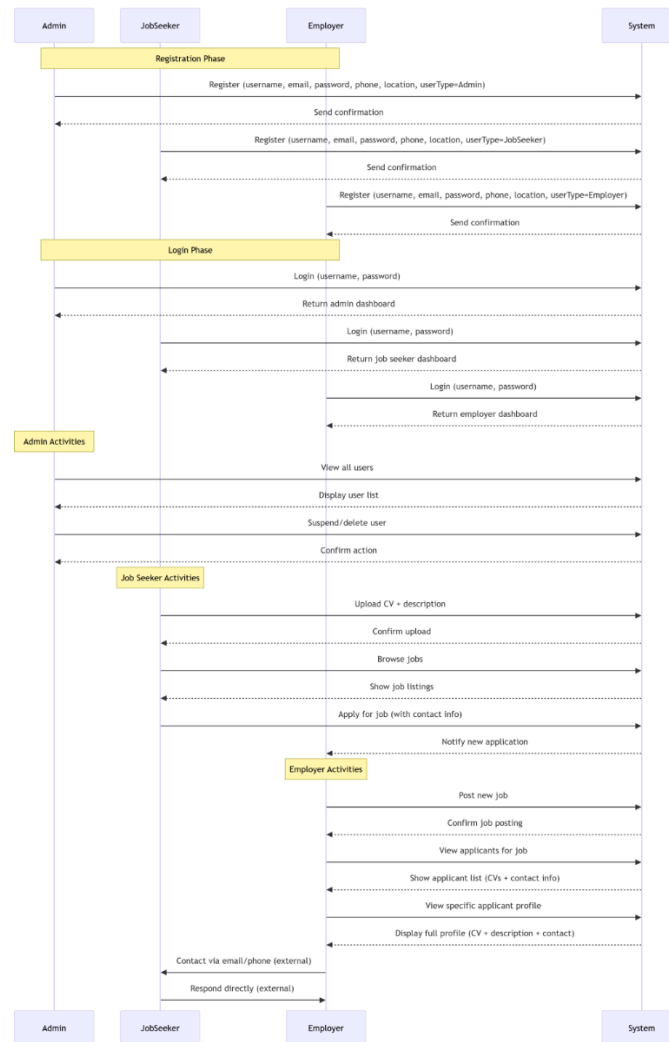


Figure 3-5 Sequence Diagram

## 3.2.2 Website Representation

### 3.2.2.1 Website design

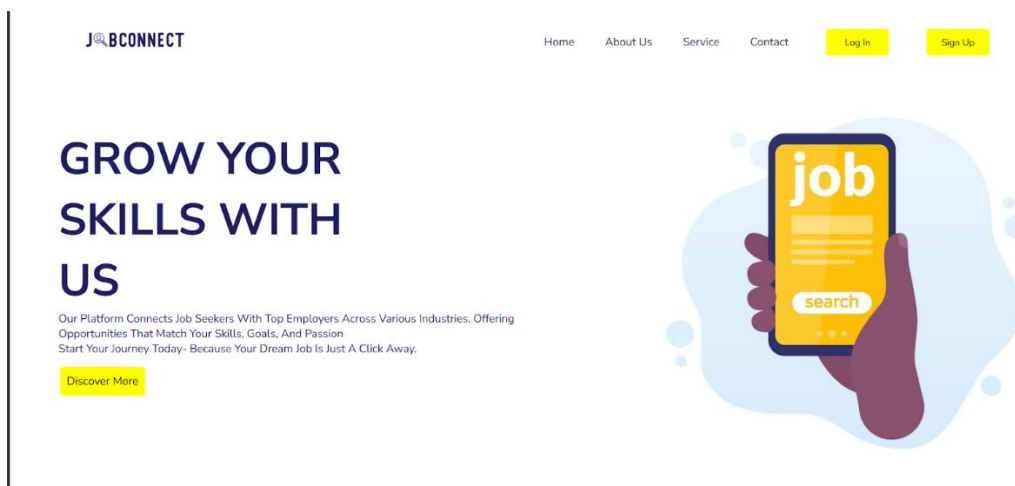


Figure 3-6 Website Design

### 3.2.2.2 Mobile Mockup

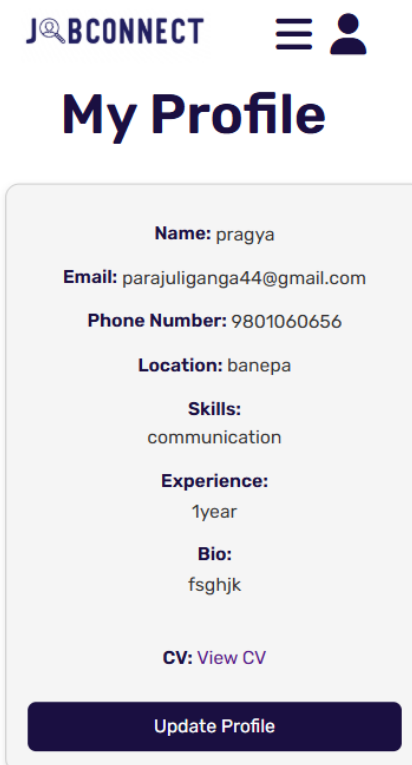


Figure 3-7 Job seeker page

**JOBCONNECT**  

## ENTER COMPANY INFORMATION

**Company Name**

**Company Description**

**Company Logo**

No file chosen

Figure 3-8 Employer Page

### 3.2.3 System Architecture:

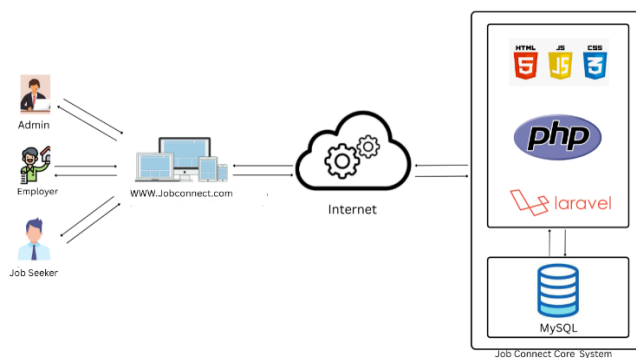


Figure 3-9 System Architecture

## 4 Chapter 4: Work done

### 4.1 Discussion on Achievements

#### 4.1.1 Challenges:

As students developing the **JobConnect** platform, we encountered several technical and organizational difficulties throughout the process. Our limited experience with full-stack development presented a steep learning curve. Understanding the architecture of a multi-role system (Admin, Employer, Job Seeker) took time, as it required mastering backend integration, session control, and database design.

We also struggled with coordinating team tasks and schedules. Balancing academic workloads and project meetings often led to delays. Despite these obstacles, we continued to make progress by supporting one another and gradually improving our technical skills through testing, and teamwork.

#### 4.1.2 Features:

This section outlines the major functionalities that have been successfully implemented in the "JobConnect" web platform. The website is designed to facilitate interaction between job seekers and employers, with administrative oversight for user and content management.

##### 4.1.2.1 User Authentication and Role-Based Access

The platform supports a secure registration and login system for three user types: Admin, Employer, and Job Seeker. Each user type is authenticated and redirected to their respective dashboards upon successful login. New users can register, while existing users can securely access their accounts using login credentials.

If an Admin deletes a user, that user is automatically blacklisted. Blacklisted users are denied future login access, effectively ensuring platform security and integrity.

##### 4.1.2.2 Admin Functionalities

Once logged in, the Admin is directed to an overview dashboard displaying critical system metrics, including:

- Total number of registered users
- Total employers
- Total job seekers
- Number of blacklisted users

The Admin can manage platform users through the "Manage Users" section. From here, they can delete users, which results in those users being blacklisted and barred from logging in again. Additionally, Admins can view user-submitted feedback, helping them monitor platform performance and user experience.

#### 4.1.2.3 Employer Functionalities

Employers are directed to the Employer Page after login. Here, they are required to submit essential company details, including:

- Company Name
- Company Description
- Company Logo

After submitting company information, employers gain access to job posting features. They can:

- Create job postings with specific requirements
- Define job status and expiry date
- View, update, or delete the jobs they have posted

Employers can also view a list of applicants who applied for specific jobs. For each applicant, they can:

- View the CV
- Update application status (e.g., selected, rejected)
- Contact applicants via email directly from the platform

Additionally, employers can browse through job seekers who have not applied to any of their job postings. By reviewing their CVs, employers can proactively reach out to suitable candidates.

#### 4.1.2.4 Job Seeker Functionalities

Job seekers, upon login, are taken to their personal dashboard. Before applying for any jobs, they are required to upload their CV. Once uploaded, job seekers can:

- Browse available job listings
- Apply to active (non-expired) job posts
- View company logos for a branded experience
- Update their personal profile information

Furthermore, job seekers can track all the jobs they have applied for under the “My Applications” section.

## 4.2 Activity Matrix:

| Week             | Activity Planned                                            | Activity Achieved                                                                                                  | Deviation and Causes |
|------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------|
| 01 May-07may     | Discuss more about website we are creating and divided work | Started Frontend and design.                                                                                       | Internal Exams.      |
| 08May-14 May     | To develop necessary database.                              | Developed database table.                                                                                          |                      |
| 15-21May         | To implement the email-verification                         | Developed registration and login functionality with email verification                                             |                      |
| 22-28 May        | Develop the employer and jobseeker landing pages.           | Developed the landing pages for employer and job-seeker.                                                           |                      |
| 29 May-04 June   | Enable CV re-upload, replacing the old file automatically   | Job applications now require an uploaded CV. Users can update their CV anytime—new uploads overwrite the old file. |                      |
| 05June-11 June   | Discuss to connect frontend and backend.                    | Connected the front-end and back-end.                                                                              |                      |
| 12 June- 18 June | To complete the function of each user.                      | Completed the function of each user.                                                                               |                      |
| 19 June-25 June  | Added the expiry option for employer                        | Added the expiry option and if the job is expired jobseeker can't apply for job                                    |                      |
| 26 June- 02 July | Preparation of Progress report, and presentation.           | Prepared Report and presentation.                                                                                  |                      |

Table 1 Activity Matrix

## 5 Chapter 5: Works to be done

### 5.1 Limitation:

While developing *JobConnect*, we focused on building a functional platform that connects employers with job seekers. However, due to time constraints and our current level of experience, there are a few areas where the system still has limitations:

- **No Instant Notifications**  
Right now, users don't receive real-time notifications. For example, if a job seeker gets selected or if a new job is posted, they have to log in and check manually. This can cause delays in communication and missed opportunities.
- **Manual CV Review**  
Employers have to go through each CV by themselves, as there's no system to filter or recommend candidates automatically. For now, it's a basic approach, which works, but might be time-consuming when the number of applicants grows.
- **Expired Job Posts Stay Visible**  
When a job expires, job seekers can no longer apply, which is good. However, employers aren't reminded or prompted to update or remove those posts, so some old listings might remain unless they're manually handled.
- **No Built-in Messaging**  
Employers can contact job seekers via email, but there's no messaging system inside the platform itself. As a result, communication takes place outside the site and isn't tracked within the system.

While these limitations highlight areas where our website falls short, they also present opportunities for future improvements. Addressing these gaps would greatly enhance the both users experience and increase the potential for this platform to succeed in the competitive market.

### 5.2 Future Work:

This project opens the door for exciting improvements in future versions of *JobConnect*. These additions will make the platform smarter, more efficient, and easier to use for all types of users.

#### 5.2.1 Immediate future works:

- **AI-Based CV Screening**  
As the number of job seekers grows, it becomes difficult for employers to go through every CV manually. To solve this, we plan to integrate artificial intelligence that can automatically scan, analyze, and rank CVs based on the job requirements. This will save time for employers and help them quickly find the most suitable candidates.

#### 5.2.2 Future works that can be done after Completion of project

- **Real-Time Notifications**  
In future versions, we aim to implement real-time alerts for key activities—such as application updates, new job posts, or status changes. This would keep users informed without having to

constantly check their dashboard.

- **In-Platform Messaging System**

Currently, communication between employers and job seekers happens through external email. We plan to introduce a built-in messaging system so that users can easily connect, ask questions, and follow up—all from within the platform.

Incorporating these improvements would make the platform more user-friendly, efficient, and adaptable, giving it the potential to succeed in the competitive online job portal space.

## **6 Chapter 6: Conclusions**

### **6.1 Conclusions**

Developing JobConnect has been a valuable and rewarding journey that helped us understand both the technical and human aspects of building a job portal tailored for local communities. Despite facing challenges with technology integration and teamwork coordination, we successfully created a functional platform that bridges the gap between employers and job seekers, especially at the grassroots level.

Our system supports multiple user roles with secure authentication, streamlined job posting, and application management features, addressing key pain points experienced by small businesses and individuals seeking employment. While some features like instant notifications and AI-based CV screening remain to be developed, the foundation we built offers a strong starting point for future improvements.

JobConnect focus on local employment and skill-based matching makes it stand out from existing platforms, promoting economic growth and inclusivity within communities. We believe that with continued enhancements and user engagement, this platform has the potential to significantly simplify the hiring process and empower users to find meaningful job connections.

Overall, this project has strengthened our technical skills, deepened our understanding of user-centered design, and highlighted the importance of collaboration in software development. We look forward to further refining JobConnect and contributing to the evolving landscape of digital employment solutions.



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